DSA Syllabus

1. Brute Force and Implementation Techniques

Introduction to brute force approaches for problem-solving. Understanding how to design straightforward solutions, convert logic into code, and analyze their limitations. Emphasis on identifying patterns and optimizing basic implementations.

2. Time and Space Complexity Analysis

Study of algorithm efficiency using Big-O,notations. Analysis of best, average, and worst-case scenarios. Understanding trade-offs between time and memory usage through practical examples.

3. Quadratic Sorting Algorithms

Detailed study of basic comparison-based sorting algorithms with quadratic time complexity. Focus on understanding internal working, use cases, and performance limitations.

- **Bubble Sort**: Concept, algorithm, optimization techniques, and time-space complexity analysis.

- **Selection Sort**: Algorithm design, step-by-step execution, and comparative analysis with other sorting techniques.

4. Binary Search

Efficient searching technique for sorted data. Understanding iterative and recursive implementations, preconditions, and time complexity analysis.

5. Recursion

Concept of recursion, base and recursive cases, recursion stack, and problem-solving using recursive approaches. Applications in searching, sorting, and mathematical problems.

6. Object-Oriented Programming (OOP) Concepts

Introduction to core OOP principles such as classes, objects, encapsulation, inheritance, polymorphism, and abstraction. Applying OOP concepts while solving data structure problems.

7. Stack

Introduction to stack data structure and its operations (push, pop, peek). Implementation using arrays and linked lists. Applications such as expression evaluation, recursion handling, and undo/redo operations.

8. Merge Sort

Divide-and-conquer based sorting algorithm. Understanding recursive splitting, merging process, stability, and time-space complexity analysis.

9. Linked List

Concepts of singly linked lists. Operations including insertion, deletion, traversal, and searching. Comparison with arrays and practical applications.

10. Binary Trees

Introduction to tree data structures with focus on binary trees. Understanding tree terminology, traversals (inorder, preorder, postorder), and basic operations.